Primary Election Analysis: Endorsements & Demographics

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Spring 2025

Data 102

1. Introduction & Research Questions

Crucial to democratic societies, elections allow citizens the opportunity to exercise their right to vote; they consider candidates who will best represent their communities and strongly reflect their values and needs, while also empowering individuals to address critical topics that affect their daily lives such as healthcare, education, and climate health, and to make well-informed decisions.

Not only are elections pertinent for representation, but it empowers individuals to be more engaged with societal issues, upholds accountability in public offices, and enforces the responsibility of the government to remain for, of, and by the people.

This study delves into the dynamics behind electoral processes by focusing on two questions:

1) Does receiving an endorsement from a political figure or organization boost a candidate's vote share?

2) Can a candidate's demographic characteristics predict the outcome of their primary (i.e.win/lose)?

Our first question utilizes causal inference to determine whether receiving an endorsement from a recognized individual or organization directly increases a candidate's vote share in a primary election, or whether the two attributes are merely correlated.

Causal inference allows us to focus on the direct impact of endorsements, while controlling for confounding variables. Because our data is observational, it introduces limitations to our method as it excludes us from unobservable counterfactuals and all confounders.

To address this, we will use a multiple linear regression model to control for certain variables (e.g. incumbency, race, gender, opponents) and estimate the Average Treatment Effect (ATE) of receiving an endorsement on a vote share. Understanding the impact that endorsements can have on an election outcome can influence the weight that candidates place on garnering endorsements towards their campaign.

Our second question explores whether a candidate's demographic characteristics (e.g. race, gender, and incumbency status) can predict the outcome of their primary election.

We will use a Generalized Linear Model (GLM), more specifically, a logistic regression model to interpret the data as it is well-suited for binary classification (i.e. win or lose). Because logistic regression assumes a linear relationship in the log-odds, we will leverage a Random Forest classifier to capture the complex, nonlinear relationships that were inaccessible through the GLM.

A limitation of a Random Forest is its reduced interpretability in comparison to GLMs, which offer concise coefficients that quantify the impact of each variable. Nonetheless, using both methods enables us to assess both linear and nonlinear patterns between demographic variables and election outcomes.

Our findings could uncover systemic biases that are hidden within voting decisions. They can be used to encourage voters to make informed decisions based on qualifications rather than a demographic characteristic. Moreover, these findings can aid in recognizing the impact that demographic features can have on a campaign.

2. Prior Work

Similar to our first research question, Shieh et al. aims to evaluate the causal impact of endorsements on electoral outcomes in their paper, *"Campaign Endorsements and Election Outcomes in Judicial Elections"*.

While we focus on endorsements made in the primary U.S. elections, they focus on how endorsements affect success in judicial elections, specifically the outcomes from different types of endorsements (e.g. from newspapers, legal associations) and the level of likelihood that are associated with such.

However, the overarching theme of endorsements' impact is still shared. The authors addressed their research by conducting a regression analysis with the dependent variable as vote share while the independent variables were the presence of endorsements, the characteristics of the opponents, and district-level factors.

Similar to our methodology, the use of regression allows us to control for confounders. To reduce the impact of the confounders and bias more strategically, Shieh et al. incorporate coarsened exact matching (CEM). Their methodologies lead them to their key findings in which group endorsements do have a positive impact on the outcome of a candidate's campaign, whereas political endorsements and newspaper endorsements do not.

In contrast, our use of ATE may be limiting as the accuracy of our result relies on controlling for all relevant confounders, and if not done properly may lead to a biased estimate. However, to strengthen our approach we will combine regression with robustness checks to ensure that we include key confounders.

Armstrong and Andreas' paper *"Predicting Elections from Biographical Information about Candidates"* assisted us in researching for our second question: Can a candidate's demographic characteristics predict the outcome of their primary (i.e. win/lose)?

Their study looked into whether or not basic biographical characteristics of candidates, whether that be occupation, education, political experience, or demographics, can be used to predict their electoral success. This helped us in our goal of utilizing candidate-level attributes in order to forecast the primary outcomes and provided a useful comparison before coming up with our approach.

While they had a broader range of biographical information, both our studies rely on observable candidate traits to inform predictions.

Our dataset is limited to gender, race, and incumbency, but the overall structure of utilizing individual-level traits remained consistent. Similar to our research, their outcome variable was electoral success. An electoral success or failure is ultimately a binary win/loss variable, which aligns well with our use of logistic regression and random forests.

However, a difference in methodology came up as the Armstrong-Graefe study used index models and judgement bootstrapping. While these methods rely on expert-based decision rules and heuristics to weigh attributes, they are more subjective than our data-driven, algorithmic models.

Our study uses logistic regression and random forests to model the primary election outcomes based on our given limited demographic variables. Though our models are less nuanced than the expert-informed or biographically large models, our models still remain replicable for analysis.

3. Data Overview

Our data can be found through the class-provided data from the 2022 U.S. Primary Elections by FiveThirtyEight , which includes detailed information on candidates in the 2022 Primaries (Senate, House, and Governor). For our research, we utilized two datasets that can be found within the directory:

1) dem_candidates.csv: contains information about the Democratic primary candidates in 2022.

2) rep_candidates.csv: contains information about the Republican primary candidates in 2022.

The information presented within both datasets include the following attributes: *Candidate*, *Gender*, *Race 1*, *Race 2*, *Race 3*, *Incumbent*, *Incumbent Challenger*, *State*, *Primary Date*, *Office*, *District*, *Primary Votes*, *Primary %,*, *Primary Outcome*, *Runoff Votes*, *Runoff %*, *Runoff Outcome*, *and a list of endorsers.*

The data generated in a curated census of the candidates in the 2022 U.S. Primary Elections, however, according to FiveThirtyEight, the dataset only includes candidates who appeared on the primary ballot and/or were attainable through public forums (i.e. news, websites, endorsements).

The information is also limited to candidates within the two-party system (i.e. Democrats and Republicans) and excludes any late changes to the race (e.g. candidate dropping out of race). These limitations may skew the generalizability of our models, though we will address more about methodology in a future section.

Each row in both datasets represents a single candidate in a specific primary election race in 2022, thus the granularity is at the candidate-election level. This granularity is essential to both of our research questions, as we investigate (1) the causal effect of endorsements on vote share, and (2) whether demographic and campaign characteristics can predict electoral success.

This level of detail allows us to model individual-level outcomes and control for relevant covariates. There can be a risk of selection bias as the data was manually collected by FiveThirtyEight with publicly ready information via websites, news, and campaign ads. The dataset does not capture demographic dimensions such as age, education level, socioeconomic status, or sexual orientation, limiting the richness of our identity-based analyses, which could have been useful for understanding our second research question.

However, with what was available to us, we calculated the AIC between various models including different features to determine those that could be most valuable to us.

The endorsement variables in both datasets included an extensive number of missing values, as not all candidates received an endorsement, nor did all endorsers endorse a large number of candidates. Regarding our research questions, we found the missing values to be relevant and kept them as binary values.

To better handle our first question and improve workability, we converted *Primary Votes* from a string value to a numerical value, and vote share into a proportion from 0 to 1. In doing so, the numerical data allows us to work with metrics and estimate our ATE.

For our second research question, we chose to model our findings with a logistic regression and Random Forest classifier. To train our models, we binarized the following features: gender (male

= 1), race (POC = 1), incumbent status (yes = 1), party affiliation (rep = 1), an endorsement from Trump (yes = 1), and an endorsement from Sanders (yes = 1). In doing so, we are able to interpret our coefficients meaningfully, without skewing the algebra from ordinal values (i.e. each race encoded into a different number).

4. Exploratory Data Analysis

To better understand our dataset and gain insight on the relationships between certain features and electoral outcomes, we perform Exploratory Data Analysis (EDA). We focused on demographic characteristics and whether or not candidates received at least one endorsement.

To visualize the potential relationship between endorsement and primary outcomes, we created stack bar charts for both the Republican and Democratic parties **(figure 1)**. Evidently, Republicans won significantly more often than their non-endorsed counterparts.

In contrast, among Democrats, endorsement status did not clearly correlate with winning. In some cases, non-endorsed candidates performed better. These differences suggest that endorsements can have a causal effect on primary outcomes—but in a party-dependent way. We interpret this as justification for using causal inference techniques stratified by party.

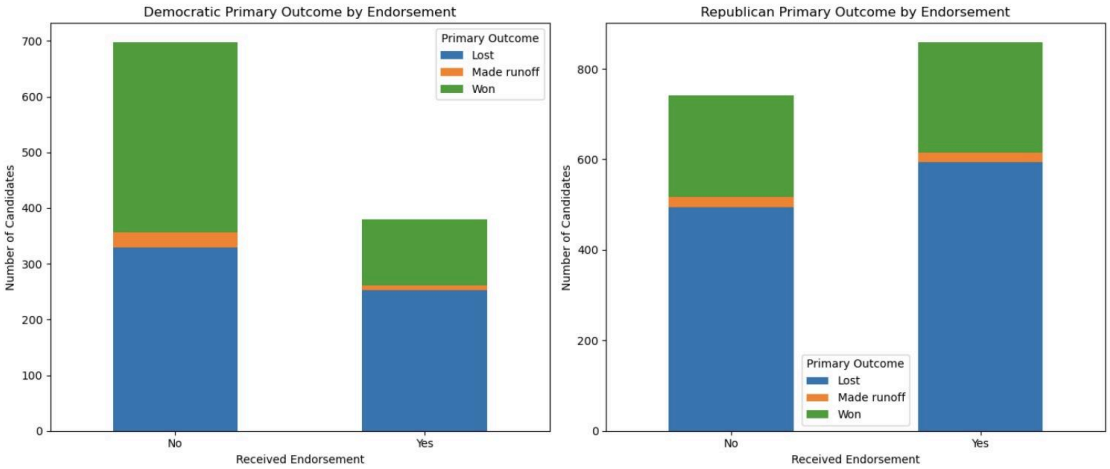


Figure 1: Party Primary Outcomes by Endorsement.

Leveraging box plots, we visualize vote share by endorsement status across party lines **(figure 2)**. Among Republican candidates, those who received endorsements had higher median vote shares and exhibited tighter interquartile ranges, suggesting that endorsements are associated with stronger outcome performance.

In contrast, among Democrats, non-endorsed candidates had slightly higher median vote shares, suggesting that endorsements might be selectively given to candidates with weaker initial support. These trends inform our first research question and suggest party-specific differences in how endorsements affect performance.

This motivates further modeling that controls for confounding factors like incumbency and candidate quality.

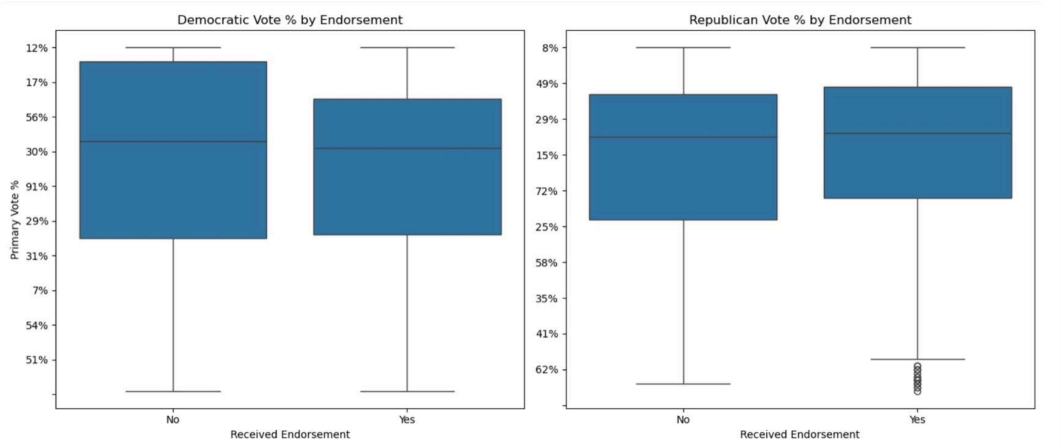


Figure 2: Box plots visualizing Party Vote % by Endorsement.

To further explore the impact that race can have on primary outcomes, we displayed the relationship between racial composition and outcome (Note: white candidates make up the majority in both parties) **(figure 3)**.

Among Democrats, white candidates were more likely to win than non-white candidates. For Republicans, the pattern is unclear, with relatively small numbers of non-white candidates.

While race appears associated with outcome, the imbalance in representation limits interpretability. Race may act more as a structural descriptor than a predictive feature in isolation.

This observation supports our intent to test demographic features in combination with other variables such as incumbency and endorsements.

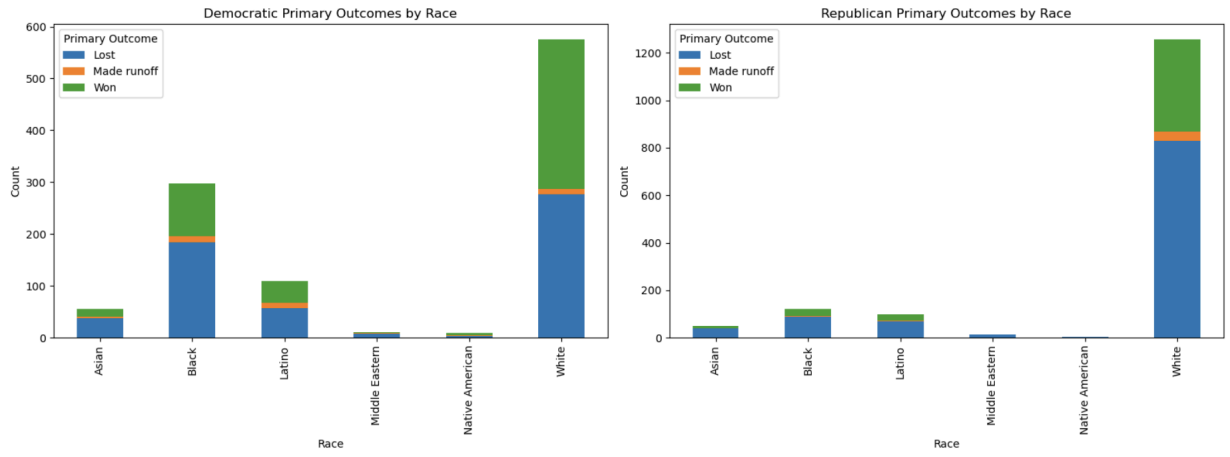


Figure 3: Primary outcomes visualized by race for both parties.

From an analysis of the proportion of primary outcomes by incumbency status **(figure 4)**, it is evident that within both parties, an exceeding majority of incumbents maintained their status following the primaries as the outcomes resulted in a win for their candidacy. This suggests

strong predictive power from the incumbency attribute, which supports the use of the feature as a variable for the second research question.

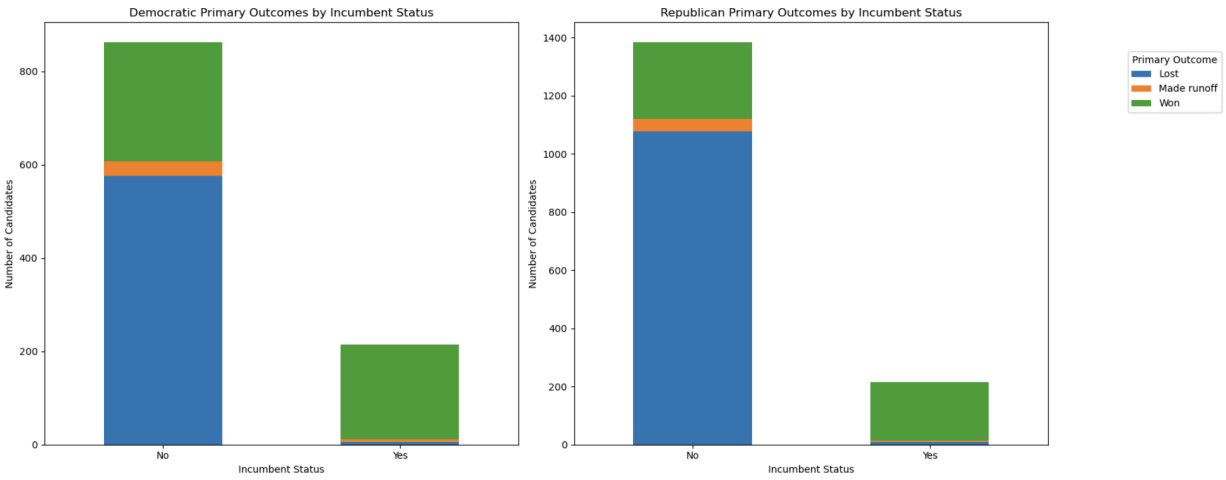


Figure 4: Primary outcomes visualized by incumbency for both parties.

To illustrate vote percentage distributions by gender within the two party system, we created box plots. In the Republican Party, male and female candidates had similar vote percentages, with overlapping distributions **(figure 5(a))**.

However, in the Democratic Party, female candidates showed higher median vote shares than male candidates **(figure 5(b))**. This motivates our second research question, suggesting that gender may have predictive power—particularly within the Democratic Party. We plan to include gender-party interaction terms in our predictive models.

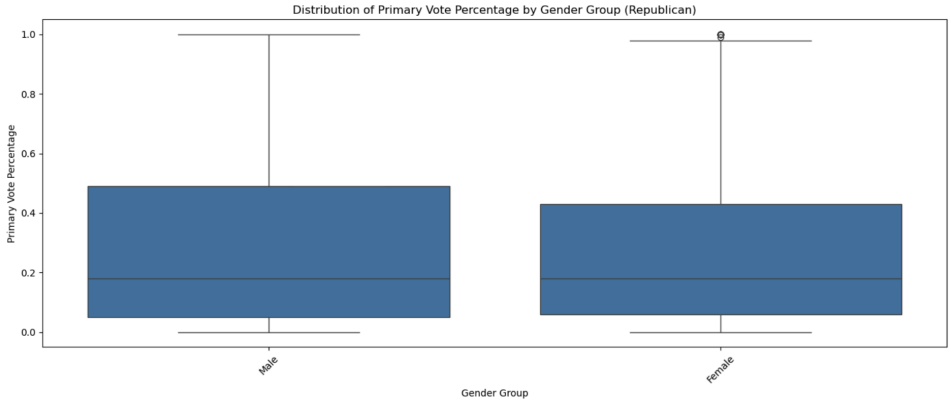


Figure 5(a): Distribution of Primary Vote % by Gender in the Republican Party.

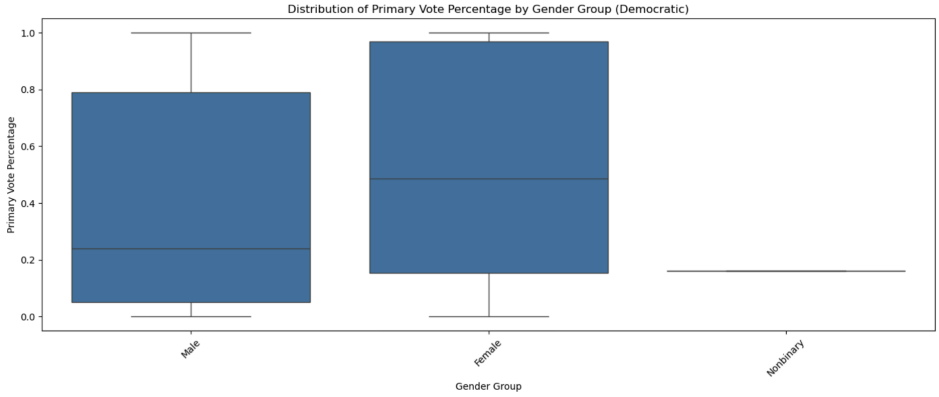


Figure 5(b): Distribution of Primary Vote % by Gender in the Democratic Party.

Looking at demographics, features such as race and gender show some influence but are not strong predictors on their own.

White candidates make up the majority, but race alone doesn't explain much variation in outcomes once we consider other factors like party. Gender shows more potential within the Democratic Party, where female candidates often perform better than male candidates, though this isn't as clear on the Republican side.

Perhaps the most consistent factor was incumbency—incumbents won at a much higher rate across both parties, which supports including it as a key control variable in our models. Overall, the EDA highlights the importance of endorsements, party, and incumbency in shaping election outcomes, while demographic variables may matter more when considered alongside other factors.

5. Research Question 1: Causal Inference

Does receiving an endorsement from a political figure or organization boost a candidate's vote share?

5.1 Methods

Our Question 1 investigates the causal effect of receiving an endorsement on a primary election candidate's vote share. Specifically, we aim to estimate the Average Treatment Effect (ATE), defined as the average difference in vote share a candidate would receive if endorsed versus if not endorsed.

The unit of analysis is the individual primary election candidate. Receiving an endorsement serves as the binary treatment, while the continuous outcome is the candidate's vote share in the primary election, scaled from 0 to 1.

In observational studies, treatment assignment (endorsement) is often not random, leading to potential confounding. To obtain a less biased estimate of the causal effect, we adjusted for pre-treatment variables that plausibly influence both a candidate's likelihood of receiving an endorsement and their potential vote share. Based on our data and domain understanding, we included the candidate's incumbency status and the type of office being sought as categorical confounders.

Our regression-based estimation relies on the assumption of unconfoundedness, which posits that, conditional on the observed confounders (Incumbent, Office), the endorsement status is independent of the potential vote share outcomes.

While including these key confounders helps to block major confounding paths, the unconfoundedness assumption is strong and relies on all relevant common causes being measured and included. Potential unmeasured confounders, such as candidate fundraising, prior political experience beyond incumbency, or inherent candidate appeal, could lead to residual confounding and bias in the ATE estimate.

We estimated the ATE using a linear regression model where the outcome (vote_share) was regressed on the treatment (endorsed) and the confounders (Incumbent, Office). The coefficient of the endorsed variable represents the estimated ATE, reflecting the average difference in vote share associated with endorsement after accounting for the included confounders.

Prior to regression, categorical confounders were transformed using one-hot encoding (sklearn.preprocessing.OneHotEncoder with sparse=False), and missing categorical values were imputed to a 'missing' category. (Numeric confounders, if used, would be scaled using sklearn.preprocessing.StandardScaler).

We considered the potential for collider bias, which can arise from conditioning on variables caused by both the treatment and a confounder. To avoid this, our set of confounders was strictly limited to variables determined before the endorsement decision.

Variables such as the primary election outcome (win/lose) or raw vote counts from the same election were not included as confounders, as they are consequences of the vote share outcome

and could introduce bias if conditioned upon. The assumed causal structure, illustrating the confounding paths blocked by our chosen confounders, can be represented by the DAG.

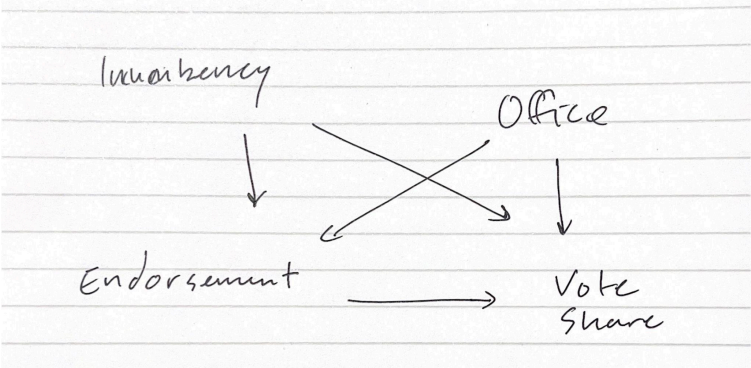


Figure 6: Causal Dag

5.2 Results

Utilizing a regression-based approach to adjust for measured confounding, our analysis included a total of 2668 complete cases out of 2676 initial candidates.

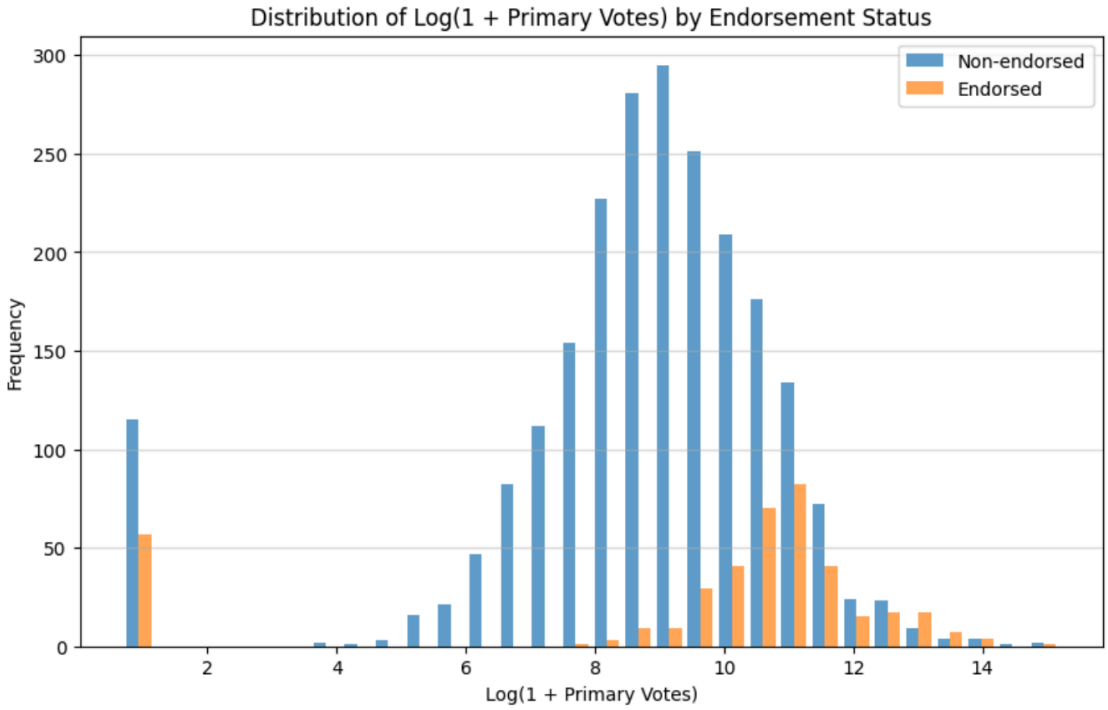


Figure 7: Distribution of Log(1 + Primary Votes) by Endorsement Status.

This figure illustrates the difference in the distribution of primary vote share (on a log scale) between candidates who received an endorsement and those who did not.

The estimated Average Treatment Effect (ATE) of receiving an endorsement was 0.1543. This suggests that, on average, receiving an endorsement is associated with an increase of approximately 15.43 percentage points in a candidate's primary election vote share, after controlling for their incumbency status and the type of office being sought.

The 95% confidence interval for this estimate is [0.1225, 0.1858]. As this interval does not contain zero, the estimated positive effect of endorsement is statistically significant at the 95% confidence level.

The confidence interval indicates that, based on our model and data, the true average causal effect is estimated to lie between a 12.25 and 18.58 percentage point increase in vote share. The confidence interval was estimated using a non-parametric bootstrap with 500 stable resamples, supporting the reliability of the interval estimate.

Balance diagnostics on the complete cases (N=2668) revealed notable differences between endorsed and non-endorsed candidates on the measured confounders, highlighting the importance of this adjustment.

For instance, 56.1% of endorsed candidates were incumbents, compared to only 9.0% of non-endorsed candidates. Including incumbency status and office type in the regression model serves to isolate the endorsement's effect from these pre-existing differences in candidate characteristics.

```
=== Final ATE Estimation (Regression) ===
Complete cases used: 2668/2676
Confounders used in model: ['Incumbent', 'Office']
ATE: 0.1543
95% CI: [0.1225, 0.1858]
```

```
=== Balance Check (on complete cases used for ATE) ===

Incumbent distribution by Endorsement Status:
Incumbent        No       Yes
endorsed
0          0.909934  0.090066
1          0.439206  0.560794

Office distribution by Endorsement Status:
Office    Governor  Representative   Senator  Senator (unexpired term)
endorsed
0         0.108168        0.772627  0.110375                 0.008830
1         0.121588        0.764268  0.111663                 0.002481
```

Figure 8: Results.

5.3 Discussion

While our regression analysis adjusted for measured confounders like candidate incumbency status and office type, the reliability of the causal estimate depends heavily on the assumption of unconfoundedness conditional on these variables.

It remains plausible that unmeasured factors, such as candidate fundraising prior to endorsement, specific prior political experience not captured by incumbency, or inherent candidate charisma, influence both who receives an endorsement and electoral success, potentially leading to residual bias.

Despite this, the statistically significant positive estimate observed (ATE = 0.1543, 95% CI: [0.1225, 0.1858]), coupled with the considerable reduction in the estimated effect size after adjusting for measured confounders compared to the unadjusted difference in means, provides moderate confidence that a genuine positive causal effect is present.

However, the possibility of unmeasured confounding means we cannot entirely rule out that some of the observed association is invalid.

To strengthen confidence in the causal claim, future research would benefit from additional data capturing these potential unmeasured confounders. Quantitative measures of candidate resources, such as pre-endorsement campaign finance data or detailed past vote totals from previous elections, could enable more robust covariate control.

Furthermore, data providing insight into the endorsement selection process itself or more granular district-level characteristics could facilitate the use of alternative causal inference methods potentially better suited to address complex selection dynamics.

Beyond a direct causal influence on voter behavior, an alternative plausible explanation for the observed positive association, even after controlling for measured factors, is that endorsements act as a signal of unmeasured candidate quality. Candidates already likely to perform well due to unmeasured traits may be more successful at obtaining endorsements, leading to an association that partly reflects this underlying strength rather than solely the endorsement's direct impact.

This analysis shares a thematic link with Shieh et al.'s study on endorsement impacts in judicial elections, as both investigate the causal effect of endorsements on electoral outcomes using regression control for confounding.

However, our context of U.S. primary elections and focus on a single binary endorsement variable differs from their examination of specific endorsement types in judicial races.

Methodologically, while both employed regression with vote share as the outcome, Shieh et al. supplemented their approach with Coarsened Exact Matching (CEM), representing a different strategy for bias reduction compared to our regression adjusted with bootstrapping.

Our finding of an overall statistically significant positive ATE of approximately 15.4 percentage points aligns partially with Shieh et al.'s finding that group endorsements had a positive impact, though our result aggregates various endorsement sources.

6. Research Question 2: GLM & Nonparametric Models

Can a candidate's demographic characteristics predict the outcome of their primary (i.e. win/lose)?

6.1 Methods

We approached our second research question with a GLM under a Bernoulli likelihood, more specifically a logistic regression with our target variable as the primary election outcome, a binary variable where 0 = lose and 1 = win.

Considering the binary outcomes, we represented our model with a logit link function, which models the log-odds of a candidate winning as a linear function of the features (under the assumption that the predictors contribute additively and independently).

We opted for a frequentist approach with L2 regularization rather than introducing a Bayesian prior for the structural integrity of our model coefficients and to avoid overfitting. Adding priors would have been subjective and weakly informative, which defeats the purpose of a Bayesian approach.

To determine our demographic characteristics, we referred to our EDA in which **figures 4-6b** demonstrate a notable significance between race and election outcome, incumbency and election outcome, and gender and election outcome – all of which visualized by party affiliation (i.e. Democrat or Republican).

We also wanted to discover the relevance between endorsement and election outcome, but to avoid overfitting, we chose to focus on political figures – Trump and Sanders – instead of political organizations. Thus, our feature set included the following variables: gender (binarized where 1 = male), race (binarized where 1 = POC), incumbency status, party affiliation, and an endorsement from Trump or Sanders.

To handle the potential underperformance of the GLM, we utilized a Random Forest classifier, as it bodes well to nonlinear relationships and offers flexibility with the complex interactions that may be evident between our variables. Random Forests are robust to outliers and multicollinearity, which is especially useful when addressing our research question as we wanted to maximize the predictive power of our demographic characteristics and minimize the fragility that would result from outliers and noise.

Due to the binary nature of our outcome, we evaluated our model performance using classification metrics: accuracy, precision, recall, F1-score, the area under the ROC curve (AUC), and feature importance. To test our model, we used a 70/30 train/test split as our dataset was fairly small (~2500 observations).

6.2 Results

Our logistic regression model **(figure 9(a))** reached an accuracy of 81% on the test set, a fairly high percentage, though a tradeoff when noticing the other metrics. With a recall of 0.98 on losses, it demonstrates strength in identifying candidates who lost their primaries. Whereas, a recall of 0.49 on wins exhibits a weakness in identifying winners.

More suggestively, the F1-scores of 0.87 and 0.64 for losers and winners, respectively, indicate that our model outperforms in identifying losing candidates based on demographics. An F1-score of 0.64 for winners indicates that our model is only able to capture a little over half of the true winners, evidenced through the moderate recall value of 0.49.

With an AUC of 0.804, we can interpret this as our model having an approximately 80.4% chance of ranking a randomly chosen winner higher than loser in terms of predicted probability, reflecting a moderately high level of certainty in classification. Our final GLM model can be represented as such

$$P(y = 1) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_5 x_5)}}$$

where β_0 is the intercept term, and $\beta_i x_i$ represents the coefficient i for predictor i.

```
=== Logistic Regression ===
              precision    recall  f1-score   support

           0       0.78      0.98      0.87       501
           1       0.94      0.49      0.64       278

    accuracy                           0.81       779
   macro avg       0.86      0.74      0.75       779
weighted avg       0.83      0.81      0.79       779

AUC: 0.8043445483134452
```

Figure 9(a): Summary of Logistic Regression results.

From our Random Forest (RF) **(figure 9(b))**, incumbency and an endorsement from Trump were the 2 most predictive predictors with coefficients of 0.676 and 0.190, respectively. The remaining characteristics – gender, race, party affiliation, and an endorsement from Sanders — yielded marginal results in contrast.

With an accuracy of 82% on the test set, the classifier suggests that the demographic characteristics carry a moderately strong predictive power. However, with imbalanced classes – more cases of loss than win – the classification metrics offer more insight. With a recall for class 1 (i.e. winners), the RF did not exhibit significance in correctly identifying winners; for class 0 (i.e. losers), it demonstrated strength in identifying losers.

Similar to the logistic regression, the F1-scores of 0.87 and 0.68 for losing and winning candidates, respectively, signifies the model's greater success at classifying losses. Despite a higher accuracy, the RF has marginally weaker overall classification confidence at 0.783.

```
=== Random Forest ===
              precision    recall  f1-score   support

           0       0.80      0.96      0.87       501
           1       0.90      0.55      0.68       278

    accuracy                           0.82       779
   macro avg       0.85      0.76      0.78       779
weighted avg       0.83      0.82      0.80       779

AUC: 0.7832500466692516
gender_male: 0.025
person_of_color: 0.027
incumbent: 0.676
party_rep: 0.072
trump_endorsed: 0.190
sanders_endorsed: 0.010
```

Figure 9(b): Summary of Random Forest results.

6.3 Discussion

With a recall of 0.55 for winner classification, the Random Forest had a slightly higher accuracy than the logistic regression (recall of 0.49), meaning that the Random Forest is a better prediction for winners.

However, with a slightly higher AUC **(figure 10)**, the logistic regression has a stronger overall classification confidence (~ 2.1% greater). Because we used a 70/30 train/test split, we were able to preserve class balance, seeing as we had a disparity between the number of winning candidates and number of losing candidates, and ensure efficient data in both the training and testing sets.

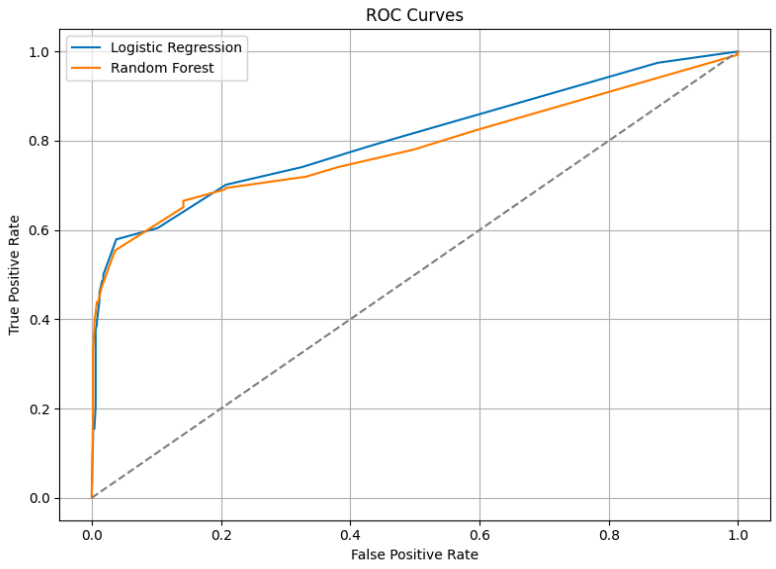


Figure 10: ROC Curves for Logistic Regression and Random Forest.

In discussion of how we chose our demographic characteristics, we followed a sequence of steps: made assumptions, visualized those assumptions through our EDA, and computed Akaike

Information Criterions (AIC) and compared them to various potential models of different features.

From our EDA **(figure 4)**, incumbency was found to demonstrate significant influence on electoral success. Similarly while gender and race had weaker relationships, they still had measurable effects, especially prevalent with male and white candidates. Moreover, we were interested to observe whether or not male/female candidates face systemic advantages and/or disadvantages, and if racial biases were evident.

Additionally, our EDA was done on a party-affiliated basis (i.e. Republican or Democrat), which we brainstormed could have weight given that ideologies and values are key to elections. Endorsements by certain political figures or organizations could also impact the outcome for candidates, following a similar reasoning of party affiliation: with the right endorsement, voters may be more inclined to vote one way, affecting the primary outcomes.

Rather than taking our features at face value and assumptions, we wanted to ensure that including all the demographic characteristics would model better than if we had excluded certain ones. To compare the quality of the potential models, we calculated the AIC values on (1) all features, (2) incumbency and endorsements from Trump and Sanders, and (3) all features without endorsements **(figure 11)**.

With an AIC of 2373.80, the lowest of the three models, we compared the differences in values to the other two potential models. We found that the differences far exceeded the threshold of 10, indicating that excluding features like gender, race, or endorsements disrupts the model quality.

```
All features AIC: 2373.80
Incumbent + Endorsement only AIC: 2443.57
No endorsements AIC: 2466.31

AIC Differences:
All vs Inc+End: 69.77
All vs No Endorsements: 92.51
```

Figure 11: AIC measures and differences.

As previously mentioned, we utilized our AUC values as the goodness-of-fit measures for our models. With the logistic regression model having an AUC of 0.804 and the Random Forest classification having an AUC of 0.783, neither model achieved perfect separation (AUC = 1.0), which limited the predictive power of the demographic features alone (i.e. these features alone cannot accurately predict a candidate's election outcome).

Importantly, AUC is a comparative evaluation metric, useful to compare models' performances, though does not directly account for overfitting. We avoided overfitting through our choices in GLM and nonparametric models, where we assumed linearity in the log-odds and independence between our observations (i.e. each candidate).

Our logistic regression model uncertainty is moderate and could stem from the variability in the model performance across the two outcome classes. With a relatively low recall for winners at 0.49 compared to losers at 0.98, this suggests that the model may be underfitting one one class, possibly due to the imbalance of data available. This indicates potential bias in predicting the less frequent positive outcomes, thus identifying losing candidates boded well with our model in contrast to winning candidates.

While a Bayesian model could make this uncertainty more explicit through posterior probabilities, we opted for a frequentist approach due to its interpretability. In contrast, with a lower AUC (0.783), the Random Forest handled the class imbalance more effectively, which was expected as Random Forests are more flexible in handling complex and nonlinear relationships within data.

Limitations include any unobservable confounders, such as campaign financing, candidate experience prior to the primary election, a candidate's media exposure and campaigning. More features could have been used to improve our models' performances and reduce uncertainty, however, this could also introduce more noise/overfitting.

We did try more interactive features (e.g. whether or not the candidate was a White male endorsed by Trump), but they added miniscule weight (<0.01) to the models as they were overtly specific, thus adding unnecessary noise. Similarly, noise that is evident in our model could be attributed to the use of multiple features. We ultimately chose to include the 6 features we had to determine the weight between combinations of predictors and to better understand our research question.

Our findings suggest that demographic features (i.e. gender, race, party affiliation, an endorsement from Trump, and endorsement from Sanders, and incumbency) hold predictive power, but they are not sufficient enough to serve as accurate election predictors.

In reference to our prior work by Armstrong and Andreas, demographic features can aid in predicting a candidate's election outcome. However, what differs between our findings and theirs is that their feature set is 10x as large as ours. Future work could incorporate more granular candidate, geographic, and financial data to refine predictions.

7. Conclusions

7.1 Outcomes & Critical Evaluation

Outcomes: Our analyses yielded great insights into primary election dynamics. In research question 1, we discovered that receiving an endorsement can have a statistically significant positive causal effect on a candidate's primary vote share.

After adjusting for our confounders: incumbency status and office, the calculated ATE was 0.1543 and a 95% confidence interval [0.1225, 0.1858]. These metrics suggest that endorsements increase vote share by roughly 15.4% on average.

For our second research question, our logistic regression achieved an accuracy of approximately 81% with an AUC or 0.804, indicating moderate classification confidence. In comparison, the Random Forest had a slightly higher accuracy of 82% with a recall for winners 0.06 higher (0.55 vs. 0.49) but lower AUC at 0.783. The models' performances were relatively similar.

In both models, incumbency and a Trump endorsement were the higher ranked predictive variables, while the remaining four: race, gender, party, and a Sander's endorsement had a marginal impact. Notably, both models performed significantly better at predicting losing candidates than winners, with F1-scores of 0.87 (losers) vs. 0.64 (winners) in the GLM and 0.87 vs. 0.68 in the Random Forest. This may be a result of class imbalance within the dataset.

Critical Evaluation: There are several limitations evident in the data and within our domain knowledge. In the first research question, the ATE relies on the assumption of

unconfoundedness, meaning we assume all variables that affect both endorsement and vote share are included within our data.

Regarding the second research question, the obvious class imbalance between number of winning and losing candidates may bias our results toward predicting losses, which inflates accuracy while reducing utility.

However, relevant variables such as campaign financing, candidates political background and experience, and public recognition are not made available, but could have great impact on our causal effect as they could well influence the candidate's ability to receive endorsements and could be strong predictors in their primary outcome. This residual confounding could bias our estimated ATE and weaken our GLM and RF.

In terms of domain knowledge, our understanding of the endorsement process was limited. Given the opportunity to speak to a domain expert, we would ask "What criteria(s) do political figures or organizations follow when choosing candidates to endorse?"

This clarification would be helpful in understanding what endorsements reflect, whether it be the candidate's quality as an individual or political future, or ideological alignment (both of which provide different implications for causal inference). As an example, if endorsements primarily follow public popularity, an endorsement may signal pre-existing strength within the candidate's campaign, thus weakening our causal implementation.

While recognizing these data limitations and the potential value of incorporating additional variables to strengthen our analysis, we were unable to identify or effectively integrate a readily

available secondary dataset that comprehensively captured these specific missing elements across our sample of candidates.

Accessing, cleaning, and merging disparate data sources containing this granular information within the scope of this project proved infeasible, despite acknowledging that such data would be necessary to fully address the identified issues of potential residual confounding and enhance predictive power. Therefore, while acknowledging the data's shortcomings, we proceeded with the most robust analysis possible using the available primary dataset.

Our results are heavily influenced by our model choices. For example, we approached our second question with a frequentist GLM and a Random Forest, but alternative models, such as Bayesian hierarchical model or generating index scores (as done by Armstrong and Andreas) could have yielded different results. However, because our models produced consistent results, our findings are likely robust to model changes.

Nonetheless, our models are limited by our feature set and class imbalance. Overall, it’s important to take our findings with caution. Demographic characteristics are not deterministic, and while certain traits have appeared to resemble predictive power, they cannot accurately predict outcomes.

Our findings are moderately generalizable within the context of U.S. primary elections, particularly those similar to the ones in scope of our dataset. Note that the generalizability is constrained by the 2022 election cycle and dynamics between candidates and party. We cannot confidently say that our results translate cleanly to elections held at a local or international level.

7.3 Recommendations

Our models indicated incumbency and a Trump endorsement were stronger predictors than race and gender, which had only a marginal impact. This likely reflects missing context like campaign finance, district demographics, experience, and media exposure.

Future research should incorporate these to examine race/gender effects alongside factors like district composition, fundraising, or media framing, clarifying how identity operates.

Similarly, our estimated endorsement effect relies on unconfoundedness, likely violated by missing confounders like candidate experience or public recognition. Incorporating these would enhance the causal estimate's validity and robustness. Also, investigating endorsement criteria would clarify what the observed effect represents.

We also propose a 'reverse burden of proof' approach (Davis). Instead of requiring underrepresented candidates to prove need, funding bodies/parties should proactively support them unless unnecessary. This shifts the default to support, acknowledging structural barriers. Predictive models offer insight but aren't ends in themselves.

Our goal is using data for democratic fairness: ensuring level playing fields and better reflecting representation complexities. By shifting standards, we can move beyond identifying disparities to actively addressing them for more diverse representation.

8. Citations

Armstrong, J. Scott, and Andreas Graefe. “Predicting Elections from Biographical Information about Candidates: A Test of the Index Method.” Journal of Business Research, vol. 64, no. 7, July 2011, pp. 699–706, https://doi.org/10.1016/j.jbusres.2010.08.005.

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